

Sentinel-2 Total Vegetation Layer – City and County of San Francisco, California

Abstract

This dataset represents estimated total vegetated land cover within the City and County of San Francisco, California. Vegetation was identified from Sentinel-2 multispectral satellite imagery using the Normalized Difference Vegetation Index (NDVI). The resulting raster classifies vegetated surfaces using an NDVI threshold greater than 0.20 and excludes mapped building footprints. The dataset was developed to estimate total vegetation extent throughout San Francisco. The layer identifies all photosynthetically active vegetation, including trees, shrubs, grasses, and other vegetation.

Data Sources

Satellite Imagery

Source: Sentinel-2 Harmonized Surface Reflectance

Google Earth Engine Collection ID: COPENICUS/S2_HARMONIZED

Acquisition period: March 1, 2025 – April 30, 2025

Number of images used: 14

Native spatial resolution: 10 meters (Bands 4 and 8)

Native projection: EPSG:32610 (WGS 84 / UTM Zone 10N)

Ancillary Data

San Francisco municipal boundary

San Francisco building footprint polygons

Cloud Processing

Clouds and cirrus were removed using the Sentinel-2 QA60 quality assurance band. Pixels flagged as cloud (bit 10) or cirrus (bit 11) were masked prior to NDVI calculation. The filtered image collection contained 14 images with scene-level cloud cover ranging from 0.00% to 99.997%, with a mean reported cloud cover of 35.44%. Because cloud masking was applied at the pixel level prior to compositing, these scene-level values do not represent the amount of cloud remaining in the final composite.

Processing Methodology

1. Sentinel-2 Harmonized imagery intersecting the City and County of San Francisco was filtered to the period March 1, 2025 through April 30, 2025.
2. Cloud and cirrus pixels were removed using QA60 quality assurance flags.
3. NDVI was calculated for each image using the normalized difference between Near Infrared (Band 8) and Red (Band 4).
4. A median NDVI composite was generated from all cloud-masked images.
5. Pixels with NDVI values greater than 0.20 were classified as vegetation.

6. Building footprint polygons were rasterized to a 10-meter grid and used to remove vegetation pixels coinciding with mapped buildings.
7. Vegetated area was calculated using pixel area and summarized across the San Francisco municipal boundary.
8. The final NDVI raster with building footprints masked was exported as a cloud-optimized GeoTIFF.

Classification Rule

Pixel Value	Description
NDVI > 0.20	Vegetation
NDVI ≤ 0.20	Non-vegetation

Building footprint areas were excluded from the final vegetation mask regardless of NDVI value.

Spatial Reference

Source imagery

Coordinate Reference System: EPSG:32610 (WGS 84 / UTM Zone 10N)

Pixel size: 10 meters

Exported dataset

Coordinate Reference System: EPSG:7131 (NAD 1983 (2011) San Francisco CS13)

Pixel size: 10 meters

File format: Cloud-Optimized GeoTIFF (COG)

Limitations

This dataset identifies photosynthetically active vegetation and does not distinguish between trees, shrubs, grasses, or other vegetation types. The 10-meter spatial resolution may not capture narrow vegetation features such as individual street trees or small planting strips. Mixed pixels containing both vegetation and impervious surfaces may result in generalized classifications in highly urbanized areas.

The vegetation layer represents median vegetation conditions during March–April 2025 rather than a single acquisition date. While cloud masking substantially reduces cloud contamination, residual classification errors may occur in areas affected by shadows, mixed land cover, or inaccuracies in the building footprint dataset.

Processing Environment

Google Earth Engine